

Newsletter Number 10 -- May 1985

The following meeting is on JULY 22th , same place and time.

SPECTRAVIDEED

NUMBER 1

SPECTRAVIDEO
318/328 Personal Computers
Have Been Voted TOP PC
OF THE YEAR By The NZ
Technology Conference
In The Home / Small
BUSINESS CATEGORY

Wellington SV Users Group

Secretary / Newsletter Editor

Don Stanley Ph 896379 (hm) , 746906 (work)

Treasurer

Colin Chin Ph 325656 (hm)

Committee members

Garry Clark 367872

Ross Wakelin 793948

Dennis Timmons 837200

Darryl Alison 85522 (Paraparaumu) (hm) 399510 (wk)

Specific areas of responsibility for committee members are ...
Don -- secretary & newsletter ; Colin -- Treasurer & MSX develop-
ments ; Garry -- Radio Access & public domain tape software ;
Ross -- National Committee ; Dennis -- library ; Darryl -- prog-
ram evaluation committee chairman & Kapiti users rep

IMPORTANT NOTICE

There is NO club night in June. Instead we will be celebrating our first birthday with a family day in JUNE . Details will be posted out to club members in May .

The JULY meeting will be our AGM . Please come along to this as there will be several items of club policy to vote on which will affect club organisation in the future. Club members will receive an agenda with the JUNE newsletter which will be posted out a little later in June due to the lack of a June meeting. We need people to assist on the committee. The position of secretary will become vacant . Anyone interested in filling this vacancy please contact Don ASAP ; likewise if you wish to nominate someone for this position. All other positions on the committee will be opened for nominations , however I understand that all other committee members will be standing again .

New members . . .

Welcome to ... Jim Quickenden , Ken & Aaron Shaw , Noel Jones , D Sharp , Tadeusz Wardhol , David McMurtrie , Graham Horne , John Hughes , Anne-Marie Wilkinson , H Zuydam , Carl Daine

Could any members not having received a membership card yet please let me know . Also note that expiry date for subscriptions

is printed on the address label of newsletter.

Number of Members = 138

Newsletter Contributions

If you are submitting programs to go in the newsletter they MUST satisfy the following conditions or they will NOT be printed.

- 1 ... be typed/printed on A4 paper with about 5 lines left blank at top and bottom of page and an inch at least around the sides
- 2 ... typed with a dark ribbon OR printed using emphasized print mode
- 3 ... include a description of what the program does , typed on A4 paper with example input
- 4 ... In the case of programs above 20 lines in length they should be sent on cassette , in this case there is no need for the printout as I can merge them into the newsletter. Enclose a stamped addressed envelope for return of cassette
- 5 ... line numbers should be continuous -- do a RENUM before printing the programs
- 6 ... PLEASE MAKE SURE THEY WORK . and would readers please note that I am NOT responsible for programs sent by people which do not work .
- 7 ... In the case of screen dumps from 3d plotters please indicate the function used and try to get the print as dark as possible -- not always easy I know .

All other newsletter contributions should be sent to me at P.O Box 7057 , Wellington South. They may miss being printed for a month if sent to any other box or given to anyone else. Articles should follow 1 and 2 above.

Deadline for Newsletter articles is the 30/31st of each month. Please do not send articles direct to the printer.

Discounts for members .. All discounts offered by Einstein Scientific and Microstyle Computers do NOT include hardware. Discounts may be offered on such peripherals as floppy disks, cassettes , books , furniture, printers , monitors but NOT on SpectraVideo hardware. Individual dealers may not offer all the goods above at discounted rates , inquire from the shops just what discounts are available to club members. Note that to obtain a discount on any

goods you must produce your current membership card.

Little things - - -

CDL are considering phasing out the retail of the single sided disk drives , the new price for a single double sided drive will be \$1395.00 , which compares to \$1895.00 at present for the dual single sided drives.

The SV modem will a 300/300 baud modem , there is zilch likelihood of that being changed. For these not knowing what is meant by baud , picture your cassette loading data 6 times slower and you have the speed at which the modem will transmit/receive data.

The SV728 MSX machine will be released in New Zealand at the May 9 Auckland PC show. It is hoped that the SV network system will be released at the same stage. The 728 is likely to retail for \$695 -- computer only no cassette or software is included.

Unfortunately the SV 606 MSX adapter has to plug into your computers expander slot , so the expander and adaptor are not able to be run together as they are supplied.

There is likely to be an expander with a 10megabyte hard disk and single 320k double sided drive built in released sometime soon. This is currently being used in conjunction with the SV network system .

Evidently any thoughts about releasing CPM3.0 have been scraped by SV .

New software which should be in the shops by now include 9 new basic titles --- Graphmaster/Morse Code/Painter/Jungle Jim/SpectraBrain/SpectraThink/Parajump/Supersaver/Anti Meteor .

And a new joystick , the Quickshot 4 has been released. Selling for \$29.95 this comes with 3 interchangeable handles. Be quick , there are not a lot going on sale initially , probably about 6 per retailer .

SV say they are unable to produce an assembler for Disk Basic for varying reasons . [Since this is the single most important piece of software for many users (and indeed the lack of one is holding back production of much software) we urge you to write to CDL and ask them to forward letters about assemblers to SV . Likewise our overseas readers , if you need an assembler lets make sure SV know about it ! .. ed]

Thanks to CDL for all the above information .

GAMES SCORES

Spectron	9700	Alexander Lindsay
Sector Alpha	666400	Gavin Knight
Kung Fu	38170	Taane Clark
	17690	Garry Clark (*)
S C Force	748600	Deirdre Chin
Telebunny	87810	Colin Chin
	26470	Don Stanley (+)
Omac	35360	Richard Bremford
	25700	Don Stanley (+)
Ninja	65200	Richard Bremford
Tetra	266600	Nicola Tuinman
Turboat	68390	Garth Thornton (*)
Sasa	118240	John Tuinman
Frantic Fred	11900	Don Stanley (level4)
Flipper Slip	21040	Don Stanley (level1)

(+) = started in normal level
(*) - started in arcade level
otherwise practice level or unknown

Extra Club Meetings

Hutt Valley owners ... on June 8th at Staff Cafeteria , Chemistry Division DSIR , Gracefield Road (which goes onto the Wainui Hill) starting at 1.00 until 4.00 , drop in ; ask questions ; get answers ; library and hardware will be available; hardware loaned for the afternoon by MICROSTYLE COMPUTERS, PETONE.

We are pleased to announce that as a result of a very successful meeting at Kapiti we will be holding regular meetings at Kapiti. These will be held at weekends starting on 7th July . The Kapiti users rep is Darryl , he will be co-ordinating meetings out there so if you require further information please contact Darryl. Meanwhile watch the newsletters for details of venue.

Also attempts are underway to form a Palmerston North sub-group of our group . If you are interested contact John , ph 722 Shannon .

Club Sponsors

Einstein Scientific -- 177 Willis Street , Wellington

MicroStyle Computers -- 198 Jackson Street , Petone



Program Evaluation Committee

What is it ... a 3 person committee who will evaluate your program and return comments on it to you

Why .. to provide software writers with unbiased comments on programs, particularly if you may wish to sell it later.

Who .. Darryl Alison , Andrew McMillen , Jonathan Clark

Where .. send programs on cassette PLUS DOCUMENTATION to Darryl at P.O Box 189 , Paraparaumu , or phone him with details.

and .. this committee will also provide a program assistance service. Ring Darryl if you have problems and he will put you on to the best person to try and solve it. This service will include problems with BASIC or CP/M and its components (eg DBASE2 , Wordstar , MBasic , 8080 Assembler , Pascal etc). If you want to be included as someone with particular skills who Darryl could advise members to contact please give him your name , phone number and what you can help with .

When sending programs to Darryl please include details of what the program should do , whether it is disk/cassette/either , give a name for it , and whether you intend selling it commercially later , printing it in a magazine (Bits & Bytes etc) or whatever.

BUGS BUGS BUGS

The A option of the draw command works as follows ;;

A0 , A2 --- always

A1 , A3 --- works only when followed by the D R & F parameters , any of L U E G or H will bomb this command.

Public Domain Software

1) Do NOT send us tapes. We will supply a standard tape which will get round the problem of people sending tapes which are too short.

2) The cost of tapes will be \$3.00 each and we will also have a charge of \$2.50 to cover handling and return postage. This is charged for each separate lot of copying done , not per tape.

3) Any of the PD Software may also be broadcast on Radio Access in our 5 minute slot on the first Saturday of each month.

4) The tapes will contain combinations of cassette & disk programs , some may be set up for 80 columns. Adapt them as you wish to suit yourself.

5) Send orders with payment to PDS ,P.O Box 7057 , Wgtn South, Wellington

6) Send us your public domain software for inclusion in the software library.

And we now have 28 CP/M disks full of public domain software. But be warned , a lot of this require a detailed knowledge of the CP/M bios and changing of such in order to utilise them . Some of the disks appear incomplete eg PASCAL (UCSD version) . PASCAL in particular requires a lot of modification to the BIOS and also requires disks formatted in a special way as it uses its own disk environment . Complete list (with number of disks in brackets) is

wordprocessor (2) help (2) othello (1) stoic (2) BDS-C (2)
ratfor/actor (1) disassembler (1) Z80 assembler (1) pilot
interpreter (1) Cobol-80 compiler (1) Fortran-80 (1) forth
interpreter (1) pascal (1) pascal utilities (3) adventure (2)
algol-m (2) cp/m utilities (2) stage-2 (1)

No , I do not know what half of them are and we would prefer NOT to have people send us disks as they take ages to copy . I suggest that you ask your friendly dealer to either copy what you want or let you make a copy yourself in the shop .

Some of this software is very old and much of appears to be very early versions of software still on the market eg Z80 assembler 1977 revision , Fortran-80 1980 etc .

Serial Numbers

If you want the club to keep a list of your hardware serial numbers then fill in the form elsewhere in this newsletter.

OOPSIE

OK I apologise to all you folks who loaded the upper case converter last month ... then tried to enter a z and make it a Z. The correction is to make the CP 7A into a CP 7B. Change line 50150 to read DATA FE,7B and all should be well . Also I seem to have counted the bytes wrong , the I1=37 should be I1=38 . Sorry about that .

Now to last months Basic seminar at the meeting . To convert a character to uppercase in Basic use A\$=CHR\$(ASC(INPUT\$(1))AND95).

THANKS

Theres a lot of people who work behind the scenes to help get the magazine ready each month. Heres some acknowledgement for the printers at broadcasting who are so prompt with printing when they get the newsletter , to Karen who helps me label them and staple them , to Dennis who posts them out from the hospital and Garry who races around town collecting them and returning them to me. Thanks folks .

And a word of thanks to these people who let our committee turn their residences upside down at committee meetings. Thanks again.

BASIC NOTES -- Using the SV LOGO

Try typing in this in direct mode

```
DEFUSR = &H4782 : A = USR(0)
```

The SPECTRAVIDEO LOGO will appear on the screen as it does when you start the system up. Furthermore , the screen will appear to lock up. Now type SCREEN 0 and the normal screen will reappear.

What happens here appears at first glance to be that the machine has produced graphics (the logo) without you telling it to go into graphics mode (you didn't type SCREEN 1 did you). In fact the system has changed one byte , , which puts it into screen 1 mode. This has many possibilities for headings for programs , bits of games etc .

Any graphics you like can be put on this screen, but there is a slight catch . Before you start putting your graphics on you need to add the instruction POKE &HFE3A,1 . This is because when the logo routine changed the single byte at FE3A , it also changed it back again without causing the VDP to reset for text mode. Thats why you need to type SCREEN 0 above to carry on . Its not in graph mode when the logo finishes but its not really in text mode either .

To see what can be done with the logo try the following little program which illustrates the logo , the GET/PUT graphics routines , and sprites using direct writing to the VRAM . These sprite VRAM addresses can be found in Deane Whitmores book , PEEKS & POKES FOR THE SPECTRAVIDEO , if you want to explore this method of using sprites further then buy it !

```
10 DEFINT J,X,Y
20 DIM A(477)
30 COLOR 1,1,1
40 DEFUSR=&H4782 : A=USR(0)
50 COLOR 15,4,1
60 POKE &HFE3A,1
```



```

70 GET (20,80)-(230,115),A
80 LINE (20,40)-(230,80),10,BF
90 LINE (20,110)-(230,150),10,BF
100 PUT (20,0),A,PRESET : PUT (20,150),A,PRESET :
    PUT (20,40),A,AND : PUT (20,110),A,AND
110 DATA 00011000
120 DATA 00111100
130 DATA 01111110
140 DATA 11111111
150 DATA 11111111
160 DATA 01111110
170 DATA 00111100
180 DATA 00011000
190 DATA B,p,e,c,t,r,a,v,i,d,e,o, , ,R,u,l,e,s
200 FOR J=1 TO 8 : READ A$ : B$=B$+CHR$(VAL("&b"+A$)) : NEXT
210 SPRITE$(0)=B$
220 J=&H1B01:VPOKE J+2,7
230 FOR I=1 TO 19
240   READ A$ : LOCATE 10,10*I-9 : PLAY"164s13a" : PRINT A$:
    LOCATE 240,10*I-9 : PLAY"164s13a" : PRINT A$
250 NEXT
260 FOR I=0 TO 6.28 STEP .4
270   X=128-99*SIN(I) : Y=96-40*COS(I)
280   VPOKE J-1,Y : VPOKE J,X
290 NEXT
300 GOTO 260

```

Software Competition

There are 2 separate competitions being run at present , one by Radio Access and one by us. The Radio Access one is not restricted to SV software and will be judged by Radio Access personnel. All computers which have programs broadcast on Saturday mornings are eligible for this competition. For details of prizes listen to Radio Access on Saturday mornings. This competition closes on 6th May 1985. All software submitted for this competition must be of an educational variety.

Now the competition we are running .

- 1) programs must be educational
- 2) entrys close on 20th May 1985
- 3) All SV entries received for the Radio Access competition will also be entered in this competition
- 4) First Prize

--- SECTOR ALPHA , by courtesy of MICROSTYLE COMPUTERS
 --- 2 Years subscription to Bits & Bytes
 --- 2 Years Users Group Subscription

THIS PAGE WANTS
YOUR~~S~~ Newsletter
CONTRIBUTIONS ! ! ! !



BILLETS REQUIRED

WE NEED ABOUT 30 BILLETS , PREFERABLY IN THE WGTN
REGION FOR PEOPLE ATTENDING THE NATIONAL COMMITTEE MEETING ON
25TH/26 MAY .

IF YOU CAN HELP PLEASE LET DON KNOW BY PHONE ASAP ... THANKS .

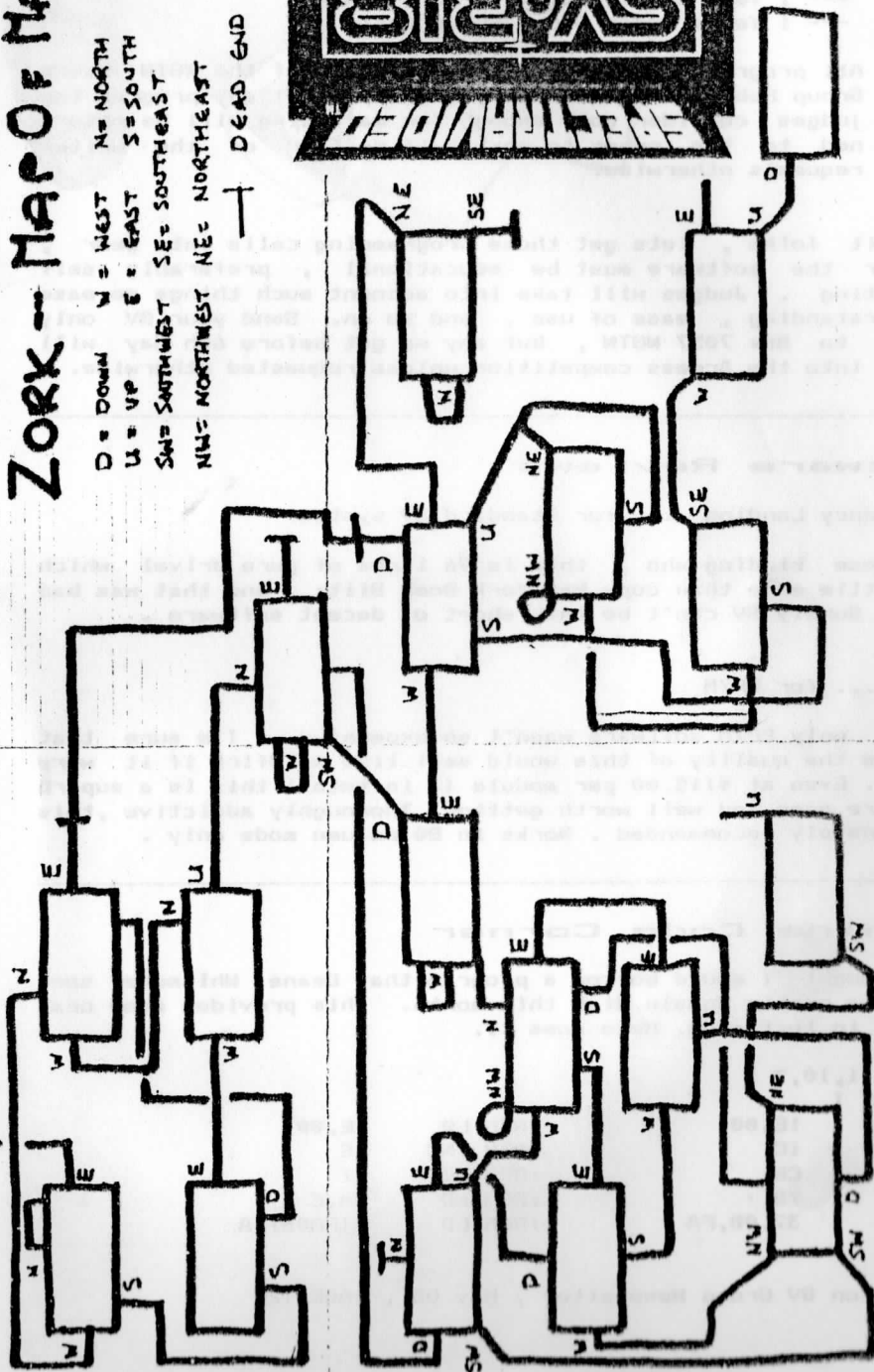
SORRY ABOUT THE SHORT NOTICE .

TROLL ROOM

ZORK -- MAP OF MAZE

D = DOWN W = WEST N = NORTH
 U = UP E = EAST S = SOUTH
 SW = SOUTHWEST SE = SOUTHEAST
 NW = NORTHWEST NE = NORTHEAST

— DEAD END



Second Prize

- \$50 discount on any purchase of \$100 or more from
EINSTEIN SCIENTIFIC
- 1 Year subscription to Bits & Bytes
- 1 Years Users Group Subscription

- 5) All programs received will become part of the WGTN Users Group Public Domain Library , except that any program the judges consider good enough for marketing will be returned to the owner to market if desired or the writer requests otherwise.

Thats it folks , lets get those programming cells into gear , remember the software must be educational , preferably self documenting . Judges will take into account such things as ease of understanding , ease of use , and so on. Send your SV only entries to Box 7057 WGTN , but any we get before 6th May will also go into the Access competition unless requested otherwise.

Software Reviews

Emergency Landing for Standard SV system

Whose kidding who , this is 96 lines of pure drivel which does little more than copy New York Bomb Blitz , and that was bad enough. Surely SV can't be this short of decent software .

ZORK ... for CP/M

If only CP/M software wasn't so expensive . I'm sure that software the quality of this would sell like wildfire if it were cheaper. Even at \$115.00 per module (3 in total) this is a superb adventure game and well worth getting. Thoroughly addictive , this is definately recommended . Works in 80 column mode only .

Machine Code Corner

I thought I would borrow a program that Deane Whitmore sent down on a public domain disk this month. This provides some neat effects in text mode. Here goes ...

2 COLOR 1,10,7

5 DEFINT I

10 DATA	1E,00	:REM LD	E,00
20 DATA	1C	:REM INC	E
30 DATA	CB	:REM RET	Z
40 DATA	7B	:REM LD	A,E
50 DATA	32,0B,FA	:REM LD	(FA0B),A

it into the memory location FA0B . Note that the A register is the only single byte register that can be used in this manner.

Why choose FA0B ? That location contains the background colour , so evidently Deane's routine is going to do something with that colour.

CALL 3750 is a machine code instruction which transfers control to the named instruction (3750) . It differs from a JP or JR in that with a CALL control is (usually) passed back to the instruction straight after the CALL . In other words , it has a similar action as Basics GOSUB .

3750 is a routine that takes the background colour and changes it , as well as the foreground and border colours. We won't go into that routine too much at present , suffice it to say that the graphic effect Deane gets from this MC routine is caused by 3750 being called 255 times in very rapid succession , with a new colour in the background each time.

Now it is a fact that machine code is very fast... sometimes its too fast and we have to slow it down to see anything happen ! The next 5 bytes , LD D,0 ; INC D JR Z,F2 are a delay , it works by constantly incrementing the D register from 0 until it overflows back around to zero again (as explained above for the E register) and then going back to the actual program which this condition (D=0) is satisfied . You could slow this down even more by using the HL register pair together , but this requires a little more work to see when the zero flag is set .

The two relative jumps at the end just go back to either the program or the delay depending on whether the delay is finished.

Sound Program ... Garry Clark

```
160 REM * 3 CHANNEL SOUND SWEEP *
170 SOUND 8,11 : SOUND 9,11 : SOUND 10,11 : SOUND 7,248
180 FOR Z=0 TO 9
190   FOR T=0 TO 255 STEP 5
200     SOUND 0,T : SOUND 1,Z
210     IF Z>1 THEN SOUND 2,T : K=Z-2 : SOUND 3,K
220     IF Z>2 THEN SOUND 4,T : L=Z-3 : SOUND 5,L
230   NEXT
240 NEXT
250 SOUND 8,0 : SOUND 9,0 : SOUND 10,0
```

Question & Answer time

Justine Pickering

1) Is there any software available to use to keep a farm cash book on the computer ...

2) Is there any way I can get my CP-80 printer to print in other than its original format straight from Just Write Jnr .

I know of no software under Basic which will solve (1) , although the SPECTRA CHECKBOOK may get near it . I suspect its not quite what you want though .

Just Write Jnr will NOT allow you to change the printer characteristics directly from the program. I am working on a program which I will make Public Domain which will allow you to read in a Just Write Jnr file and insert printer controls in it , similar to WordStar. My able assistant in this seems reluctant to let me have a tape with any JWJ file on it to work on so would someone else who owns JWJ please send me a tape with a single page or so of text from JWJ on it so I can work on this .

Survey

Please spare a few minutes to fill out the enclosed survey form which is enclosed for CLUB MEMBERS only . It will assist us to see in which direction to steer the users group .

The Keyboard Buffer

Memory locations FA1A/FA1B point to the next character to be read in the keyboard buffer. Locations FA1C/FA1D point to the last character read . When these are not the same Basic attempts to read characters until they are equal . You can clear the keyboard buffer (ie stop Basic trying read more characters) by using the statement

POKE &HFA1A,PEEK(&HFA1C) : POKE &HFA1B,PEEK(&HFA1D)

which sets the keyboard buffer so that BASIC thinks no characters need to be read.

This is very useful when using function keys . It's not unusual to have a statement like

PRESS ANY KEY TO QUIT

, if a function key defined as "COLOR 15,4,1" is pressed then the C is read by the program , which then ends , Basic finds the

other characters in the buffer , says hey lets read them , and then generates the line OLOR 15,4,1 which leaves a syntax error on the screen . BY reading the C and then resetting the buffer as above this problem is avoided .

To illustrate try this program

```
10 REM DON'T RESET THE BUFFER
20 KEY 1,"SYNTAX ERROR"+CHR$(13)
30 PRINT "PRESS A KEY TO STOP"
40 IF INKEY$="" THEN 40 ELSE END :REM PRESS F1 KEY
50 :
60 REM NOW RESET THE BUFFER
70 :
80 KEY 1,"SYNTAX ERROR"+CHR$(13)
90 PRINT "PRESS A KEY TO STOP"
100 IF INKEY$="" THEN 100 ELSE POKE &HFA1A,PEEK(&HFA1C):POKE
    &HFA1B,PEEK(&HFA1D):END :REM PRESS F1 KEY
110 :
120 REM EFFECT OF SETTING END OF BUFFER WHERE IT DON'T BELONG
130 :
140 POKE &HFA1A,20
```

When you run this program you should get a syntax error and BASIC should write YNTAX ERROR on the screen because that is what was unread from the keyboard buffer (which is 40 bytes long) .

Then rerun it using RUN 80 , you should not get the error as the buffer has been set so Basic thinks it is read completely .

Finally try RUN 140 to see the effect of pointing the buffer somewhere it should not be . Basic tries to read the whole ROM in order to match FA1A/FA1D with FA1C/FA1D , screen clears , returns , beeps and all are in there .

H A V E A N I C E D A Y

S T O P P R E S S . . .

The SX-Xpress is evidently a 64k MSX computer with a built in 3.5 inch Sony disk drive . Early production models were shown at the Las Vegas Computer Show in February . No release dates or anything else known about this beast yet

HIDDEN LINES

By Bob Curwood

If, like me, you are a compulsive reader of computer magazines, you will no doubt have come across the subject of hidden lines for (dare I say it?) other brands of computers and wondered if something similar can be done on the SPECTRAVIDEO.

The idea is that a program line (or part line) which does something significant to the program is altered in some way to prevent it being listed.

OK, here's one way to do it!
First enter the line, for example:

```
210 POKE652681,199'
```

In this example don't forget to put the `END` after the number or the computer will do it for you and upset your count.

Next count up the total number of characters in the line, inc; the `END` and the line number. For the above example it would be 19. Now append to the line that many asterisks (or whatever you like) thus giving:

```
210 POKE652681,199'*****
```

Now comes the hard part! You will have to search thru' ram, starting at `&H8001`, to find the asterisks. I suggest the following in immediate mode: (But see the last program)
`FORA=&H8001 TO 65535:THEX#(A):"CHR#(PEEK(A)):NEXT`

Use `CTRL/STOP` and note the start and finish addresses.
Next poke these locations with `CHR#(127)` like this:
`FORA=Start TO finish:POKEA,127:NEXT`

If you now list the program the line will not be seen, however as you know where it is you may see a brief flash as it appears on the VDU and is then wiped of by the deletes.

As a further refinement put the command at the end of a line, remembering to add 1 more asterisk for the `END`, and/or add 1 more asterisk which is `POKEd` with `CHR#(30)`, (cursor up).

HOW TO DISCOVER SUCH LINES!

If you suspect any hidden lines in a program they can be seen by slowing down the listing.

The easiest way to do this is with `POKE65289,118`. If this is still too fast then use the following M/C delay routine

```
10 REM Delay. Bob Curwood
20 RESTORE
30 FORA=&HD000 TO &HD013
40 READA$:POKEA,VAL("3H"+A$):NEXT
50 DATAF5,E5,21,0,40,23,7C,B5,20,F8,E1,F1,C9
60 DATA21,00,00,22,00,FF,C9
70 KEY9,"POKE65289,195"+CHR#(13)
80 KEY10,"POKE65289,201"+CHR#(13)
90 DEFUSR=&HD00D:A=USR(0)
```

Once the program has been run it can be deleted and used as follows:

To slow down the listing press `F9`

To restore press `F10`

Note that the actual delay is set by the number loaded into the HL reg; this is the 2 data values after the first 21.

SPECTRAVIDEO CLUB LIBRARY

BASIC REFERENCE MANUAL. by TOBY & ROSS.
S/V TECHNICAL REFERENCE MANUAL.
SCHEMATIC DIAGRAMS FOR :

1. S/V 318 & 328 COMPUTERS.
2. S/V 802 & 901 CENTRONICS/ S.V INTERFACE.
3. S/V 805 R.S. 232 INTERFACE.
4. S/V 803 16K RAM.
5. S/V 801 DISK CONTROLLER.
6. S/V 903 DATA CASSETTE.
7. S/V 807 64K RAM.
8. S/V 904 MONO CASSETTE.

CLUB MAGAZINES.

KIWI MSX. AUG-NOV 84.

MSX/ S.V COMPUTER CLUB MAG.
FEB, APR, JUN, AUG, OCT, DEC 84.
FEB 85.

WGTN S.V CLUB MAG.
all months

AUSTRALIAN NEWSLETTERS.
OCT, NOV, DEC 83.
JAN-OCT 84.

U.S.A. N.Z.F.O INFORMATION ON PACKET SWITCHING. DATA BASE ADDRESSES IN THE
U.S.A. N.Z.F.O INFORMATION ON PACKET SWITCHING. DATA BASE ADDRESSES IN THE
AND PRICES. TELENET DATA CHARGES INFORMATION.

ARTICLES & REVIEWS.

MSX WORLD SEPT 84.
OVERVIEW OF MSX STANDARD.

S.V.I 728.

WHICH MICRO OCT 84.
SONY H.B. 75.
SANYO M.P.C 100.

SONY HIT BIT REVIEW.

S.V.I SOFTWARE CATALOGUE.

MAGAZINES.

HOME COMPUTING MAY, JUN, JUL 84.
MSX USER. NO.1.
COMPUTER CHOICE. AUG 84.
PRACTICAL COMPUTING. JUN 84.
ELECTRONICS AND COMPUTING. AUG 84.
COMPUTING TODAY. AUG 84.
CREATIVE COMPUTING. JUL 84.
WHICH MICRO. SEP 84.
POPULAR COMPUTING. OCT 84.
YOUR COMPUTER. OCT 84.

presently 0,40. The slowest value (strange-as-it-may-seem) is 0,0 with the next slowest as FF,FF.

In case you were wondering what the example does, enter it and then try LIST or LLIST !

If you POKE65289,199 the computer will reboot, this is quicker than the usual DEFUSR=0:A=USR(0) routine.

Anyone know what the MON command is for? You can use it for a N/C routine (Instead of A=USR(0) just use MON) by placing a JP, [your routine address] at 8HFF84/5/6. E.G poke these 3 locations with C3,BE,40 which will call the BEEP routine with the MON command.

Finally, here's a 3 (or even 1) line program to find out where a program line is in RAM. If you are in the habit of numbering your programs as 1,2,3,etc (Shame on you) you will only need find room for line 3. Enter line 2 in immediate mode then GOTO 3 (or wherever it is)

```
1 'GOTO10
2 A=8H8001
3 ?"Line#";val("&H"+HEX$(PEEK(A+3))+STRING$(2-LEN(HEX$(PEEK(A+2
))) ,48)+HEX$(PEEK(A+2)))TAB(12)"is at&H"HEX$(A):A=VAL("&H"+HEX$(
PEEK(A+1))+STRING$(2-LEN(HEX$(PEEK(A))),48)+HEX$(PEEK(A))) : IFPEEK
(A)=0THEN?"Program ends at&H"HEX$(A):ENDELSE3'Curwood
```

Shew!

EINSTEIN SCIENTIFIC LIMITED
EINSTEIN SCIENTIFIC LIMITED

EINSTEIN'S have the extras for you:

PRINTERS * DISC EXPANDERS * CP/M SOFTWARE * MONITORS

The SV328 with doublesided disc expander, 80 column card and Micropro software now reduced to \$3695! Limited offer, order now.

Watch the papers and this space for details as Einstein's make 1985 Spectravideo Software Year!

Now in stock --- SpectraVideo Sound & Graphics / SpectraVideo Basic , from Sweden ; Basic Reference Guide ; SpectraVideo Computing , by Ian Sinclair

Details from Myra or Malcolm on 844 353 or 851 055.

SHOP HOURS - 177 WILLIS STREET

Monday to Thursday 9.00am-5.00pm

Friday 9.00am-7.00pm

Saturday 10am-12.30pm

EINSTEIN SCIENTIFIC LIMITED

in association with THE COMPUTER EXPERIENCE

BRANCHES at WELLINGTON * NEW PLYMOUTH * HAMILTON * PALMERSTON
NORTH * CHRISTCHURCH *

[Please note that Michael Pollard has now left Einsteins , but will be a regular attender at meetings as he is an SV owner. On behalf of the user group I would like to say THANK YOU to Michael for his assistance as an interface between us and Einstein. We look forward to continuing connection with Einstein via Myra who is the new Wellington Shop Manager .. ed]